

Verdant Coil: Core Game Design Document

One-Line Pitch

Verdant Coil is an asynchronous, biological dungeon builder where players evolve weird symbiotic creatures to crawl through user-made living mazes—and grow stronger by doing so.

Core Gameplay Loop

Player Loop (Asynchronous Model)

1. **Explore** other players' organic maze-levels ("Coils").
 2. **Gather Nutrient** (primary resource) by surviving and reaching the Heartroot.
 3. **Earn Sporeprint** (secondary resource) when others complete your levels.
 4. **Spend resources** to upgrade your crawler or expand your building toolkit.
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Player Experience

In PvE Mode (Explorer)

- Control a **symbiotic crawler** in **top-down view**.
- Navigate **organic hazards**: toxins, sticky goo, digesting tunnels.
- Movement is **tile-based**, fluid, with contextual interactions.
- Use **upgrade abilities** to pass blocked areas.

Upgrade Examples

Upgrade	Description
Hardened Skin	Survive longer in corrosive zones.
Photoreceptors	Reveal hidden signal paths or invisible modules.
Acid Sac	Digest soft walls to open secret passages.
Locomotion Shift	Pass through sticky/goo terrain.
Directional Memory	Replay previous fork paths for planning.

In Builder Mode (Creator)

- Build a **2D grid-based maze** using organic tiles.
- Place and rotate **modular biological tiles**.

- All builds must be completed by the creator before publishing.
- Build power limited by a **biomass budget**.

Tile Modules

Module	Function
Flesh Tile	Basic walkable path
Acid Pool	Damages non-armored crawlers
Sticky Web	Slows crawler movement
Pulse Gate	Requires upgrade to pass
Digest Wall	Breakable with Acid Sac
Trigger Node	Opens doors / changes map
Heartroot	Level goal tile



Resources & Progression

Resources

- **Nutrient** (primary): Earned by exploring others' Coils. Used for **upgrades**.
- **Sporeprint** (secondary): Earned when your Coils are completed. Used to unlock **new build modules** or increase biomass budget.

Validation System

- Players **must complete** their own level to publish it.
- System flags required upgrades for future explorers.
- Players can **rate** Coils after finishing them.



MVP Scope (2 Weeks)

Must-Have Features

- Top-down crawler movement
- Basic tile-based level builder
- JSON-based save/load
- Player upgrades (toggle/unlock flags)
- Exploration gameplay (hazards, goal tile)
- Level validation (must be solvable)
- Level browser + ratings

Excluded from MVP

- Cloud backend (local file storage only)
 - Animations beyond squash/stretch
 - Multiple tilesets/themes
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Expansion Potential

- Add new module types (e.g. mobile hazards, light zones)
 - New biome variants (toxic desert, techno-fungus)
 - Online level sharing and analytics
 - Cosmetic mutations / evolution skins
 - Community events and seasonal builder challenges
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Summary

Verdant Coil blends expressive player creativity with exploration-driven progression. Simple to build, rewarding to play, and rich with possibility—it's a living game world grown one Coil at a time.